

Name: _____

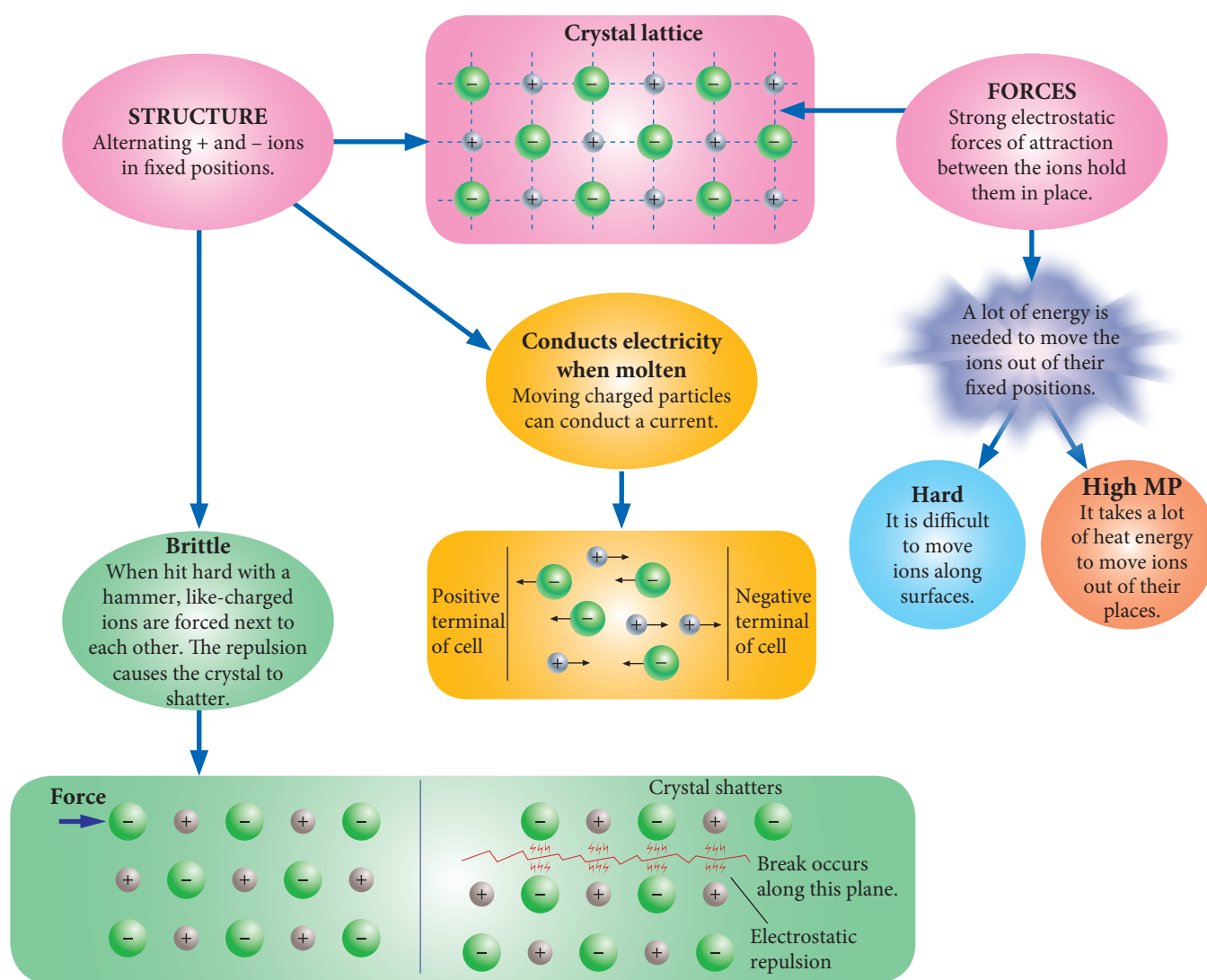
Class: _____

ACTIVITY SHEET

3.2 Interpreting and constructing summary diagrams

Part A

The summary diagram below shows how the structure of an ionic compound and the forces of attraction between ions account for the properties of those compounds.

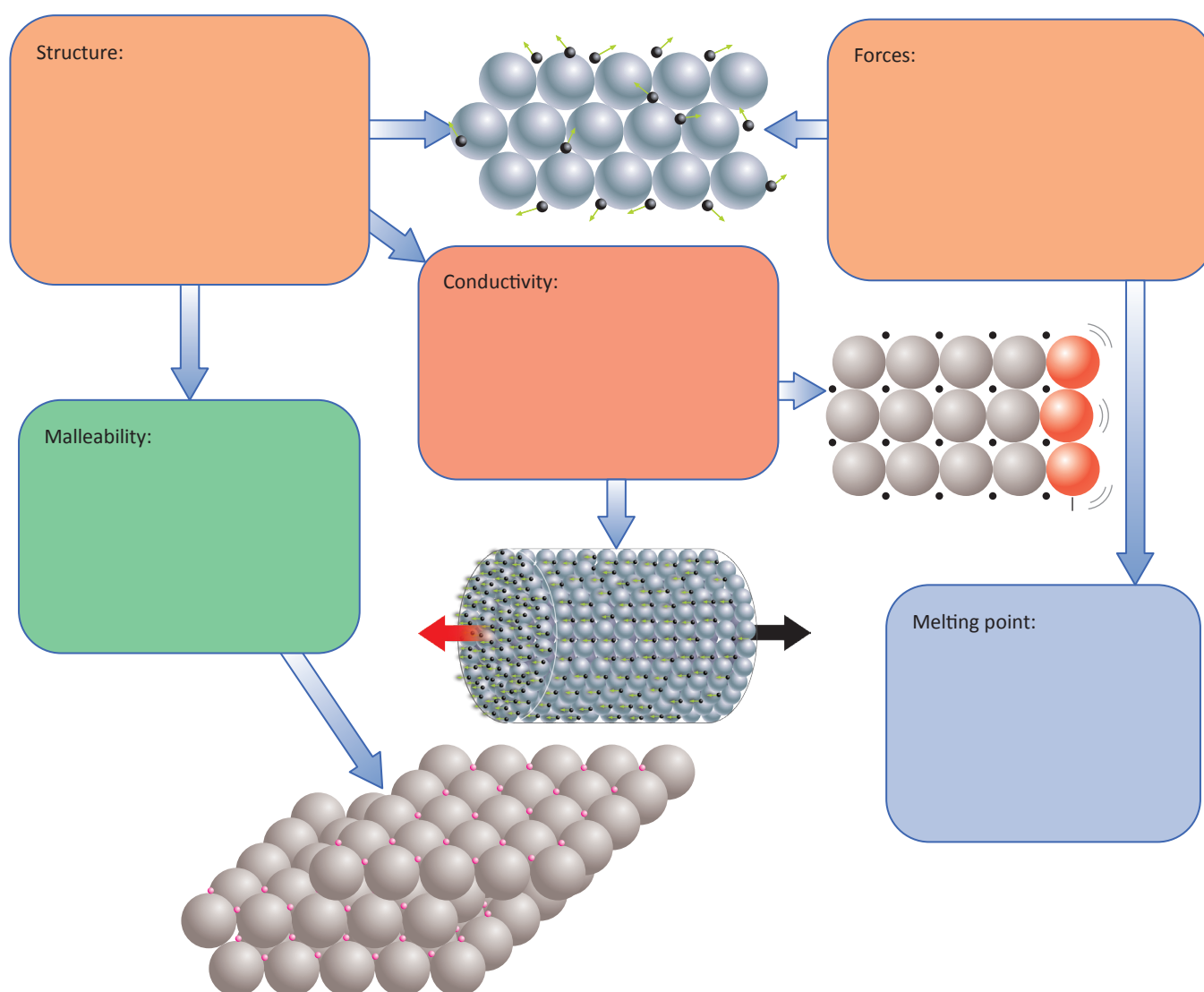


1 What keeps the positive and negative ions arranged in a regular lattice structure?

- 2 What forces occur between the ions in the lattice?
- 3 Why is it that a molten ionic compound can conduct electricity but not a solid sample?
- 4 Why do ionic crystals shatter when struck despite the strong bonds between oppositely charged ions?

Part B

Complete a similar summary diagram explaining how the structure of a metallic element and the forces of attraction involved account for the properties of a metal.



- 1 What holds together the atoms that make up a metal?

- 2 Why do metal cations occur?
- 3 What forces occur between the metal cations and the mobile electrons?
- 4 Why are metals so malleable despite the strong bonding that exists between the metal cations and the mobile electrons?